Introduction to Minimum Spanning Trees

Prim's Algorithm

Dr. Sumaiya Tabassum Nimi (STI)
Assistant Professor, Dept. of ECE — North South University

Introduction

A **Minimum Spanning Tree (MST)** of a connected, undirected, weighted graph is a subset of edges that:

- connects all vertices,
- has no cycles,
- has minimum possible total weight.

Prim's Algorithm

Prim's Algorithm grows the MST from an initial vertex, always selecting the cheapest edge that expands the tree.

Intuition

Imagine expanding a network outward from one city: you always choose the cheapest road connecting any city in the network to a new city.

Algorithm Steps

1. Select any starting node.
2. Add the smallest edge connecting a visited node to an unvisited node.
3. Repeat until all nodes are included.

A **min-priority queue** is typically used to efficiently retrieve the smallest crossing edge.

Pseudocode

Example Walkthrough

Consider this weighted graph:

Algorithm 1 Prim's Algorithm

Choose an arbitrary start vertex s.
Initialize a min-priority queue Q with edges incident on s.
Mark s as visited.
Initialize MST $\leftarrow \emptyset$.
while MST has fewer than $(n-1)$ edges **do**
    $(u, v, w) \leftarrow$ Extract-Min from Q.
    if v is not visited **then**
        Add (u, v, w) to MST.
        Mark v as visited.
        Insert all edges (v, x) into Q.
    end if
end while
return MST

Edge	Weight
$A - B$	4
$A - C$	3
$B - C$	1
$B - D$	2
$C - D$	4
$D - E$	2
$C - E$	5

Start at vertex A.

Step 1: Neighbors of A:

$$(A, C, 3), (A, B, 4)$$

Pick smallest $\rightarrow (A, C, 3)$.

Step 2: From C, consider new edges:

$$(C, B, 1), (C, D, 4), (C, E, 5)$$

Pick smallest $\rightarrow (C, B, 1)$.

Step 3: From B, consider new edges:

$$(B, D, 2)$$

Pick $(B, D, 2)$.

Step 4: From D:

$$(D, E, 2)$$

Pick $(D, E, 2)$.

Total weight:

$$3 + 1 + 2 + 2 = 8$$